Netaji Subhash Engineering College

PRESENTATION ON :- **Network Models and Protocols :-**

Computer Network (PCC-CS602)

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Network Models and Protocols: Understanding OSI & TCP/IP :-

Exploring the foundational frameworks that govern how data travels across global networks.

Why Network Models Matter :-

Networking is inherently complex, involving diverse hardware, software, and geographic distances. Network models simplify this complexity by breaking down communication into manageable, standardized stages.

Both the OSI and TCP/IP models offer crucial blueprints for understanding the intricate flow of data.

Simplify Complexity

Breaks down communication processes

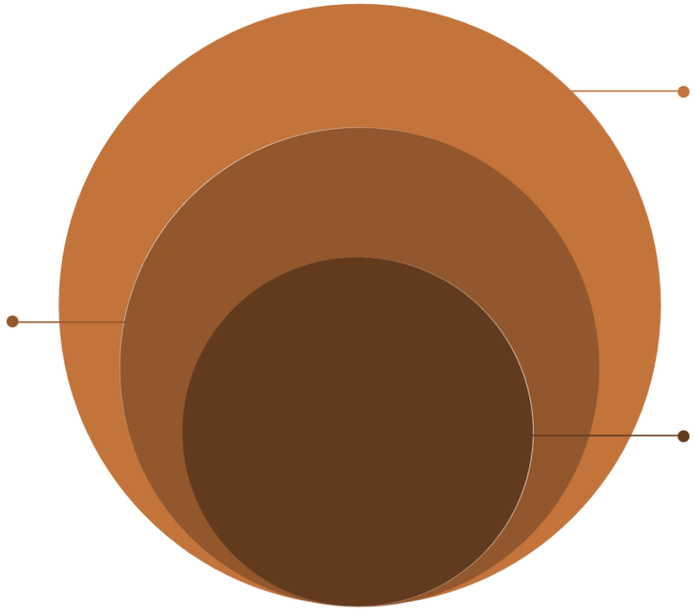
Aid Troubleshooting

Provides structured diagnostic approach

Ensure Standardization

Facilitates interoperability across devices

The OSI Model: 7 Layers of Network Communication



Developed by the International Organization for Standardization (ISO) in the 1980s, the OSI model serves as a comprehensive, theoretical framework for network communication. Each layer performs specific functions, contributing to the overall process of data exchange.

- **Application:** User interface for network services (e.g., HTTP, FTP).
- **Presentation:** Handles data formatting, encryption, and compression.
- **Session:** Manages communication sessions between applications.
- **Transport:** Ensures reliable end-to-end data delivery (e.g., TCP).
- **Network:** Responsible for logical addressing and routing (e.g., IP).
- **Data Link:** Manages physical addressing (MAC) and error detection.
- **Physical:** Defines the hardware and transmission medium (e.g., cables).

The TCP/IP Model: Practical Internet Foundation

Born from the ARPANET project in the 1970s, the TCP/IP model is the practical, widely implemented standard that underpins the entire internet. It consolidates the OSI model's seven layers into a more streamlined, four-layer architecture.

1

Application Layer

Combines OSI's Application, Presentation, and Session layers, focusing on user-facing services and data handling.

2

Transport Layer

Manages end-to-end communication, offering reliable (TCP) and connectionless (UDP) services.

3

Internet Layer

Handles logical addressing and routing of data packets across different networks using IP.

4

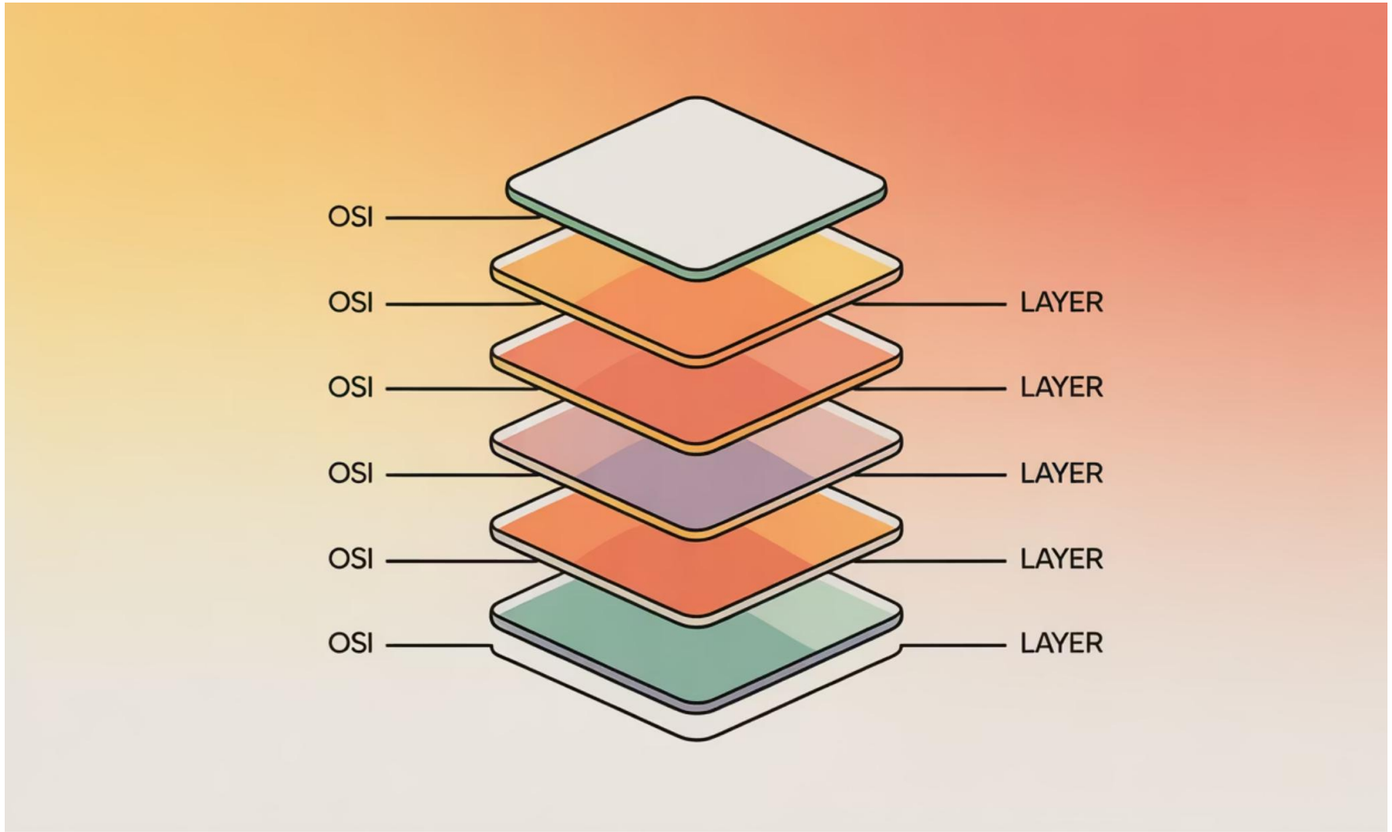
Network Access Layer

Encompasses the physical and data link functions, dealing with hardware and media access.

OSI vs TCP/IP: Key Differences

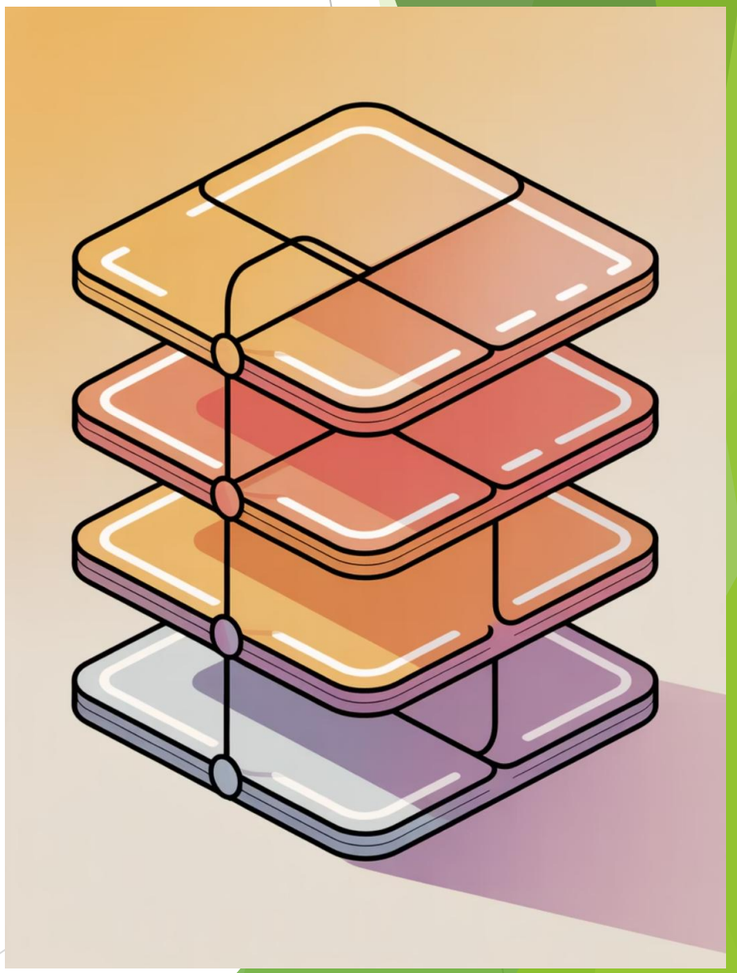
OSI Model

- *7 Layers: More granular and theoretical.*
- *Protocol-Independent: Focuses on conceptual functionality.*
- *Distinct Layers: Separates Presentation and Session layers.*
- *Clarity: Excellent for education and design clarity.*



TCP/IP Model

- *4 Layers: More consolidated and practical.*
- *Protocol Suite-Based: Directly linked to specific protocols.*
- *Merged Layers: Presentation and Session combined into Application.*
- *Interoperability: Designed for real-world implementation.*



Essential Network Protocols and Their Roles

TCP (Transmission Control Protocol)

Ensures reliable, ordered, and error-checked delivery of data packets. Crucial for applications requiring high data integrity.

IP (Internet Protocol)

Routes data packets across networks using unique IP addresses. It's the "postal service" of the internet.

HTTP (Hypertext Transfer Protocol)

The foundation of data communication for the World Wide Web, enabling browsers and servers to exchange information.

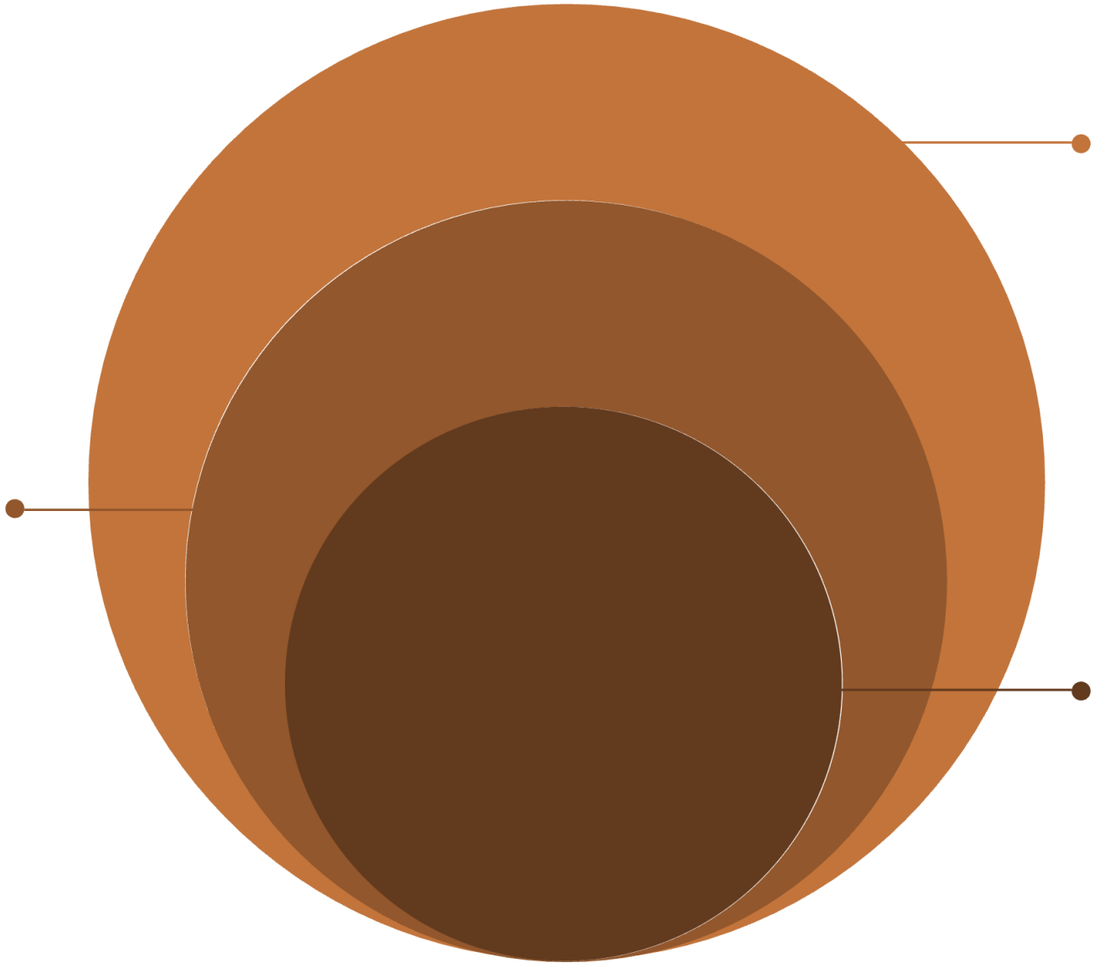
FTP (File Transfer Protocol)

Used for transferring files between a client and a server on a computer network.

UDP (User Datagram Protocol)

A faster, connectionless alternative to TCP, ideal for applications where speed is more critical than guaranteed delivery, like streaming.

How Protocols Work Together: The Protocol Stack



Outer: Network & Access

IP routing and physical bit transmission

Middle: Transport

Segments, ports, reliable delivery

Core: Application

HTTP requests and responses

The protocol stack illustrates the hierarchical nature of network models, where each layer relies on the services of the layer below it. This layered approach allows for modular development and ensures that changes in one layer do not necessarily affect others.

At the receiver, the process reverses: the Network Access layer receives bits, the Internet layer identifies and delivers packets, the Transport layer reassembles segments, and the Application layer reconstructs the original data for the user.

Summary: The Backbone of Network Communication

Understanding network models and protocols is fundamental to comprehending how modern digital communication operates. These frameworks provide the essential structure and rules for data exchange.

1. OSI Model:

   A conceptual guide for understanding layered network functions.

2. TCP/IP Model:

   The practical foundation powering the entire Internet.

3. Key Protocols:

   TCP, IP, HTTP, FTP, and UDP are vital for reliable and efficient data transmission.

4. Protocol Stack:

   Illustrates how different protocols collaborate to deliver data seamlessly.

END